

English Experiment Report: Istio Traffic Management with Bookinfo Application

1. Introduction

This experiment explores the core traffic management functions of Istio based on the Bookinfo sample application. Three key capabilities are verified in the experiment, including traffic weight distribution, user-based routing policy and service fault injection.

2. Prerequisites

- A Kubernetes cluster with Istio successfully installed
 - Bookinfo sample application fully deployed and running
 - kubectl correctly configured to connect to the Kubernetes cluster
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3. Experimental Procedures and Results

3.1 Traffic Weight Distribution (80/20 Split)

Experiment Purpose: Distribute 80% of user traffic to the reviews-v1 service without star ratings, and allocate the remaining 20% traffic to the reviews-v3 service with star ratings.

Operation Steps: Apply the traffic distribution configuration file to set the 80/20 traffic forwarding rule for different versions of the reviews service.

Experiment Result: After continuously refreshing the page, most of the review contents are plain text from reviews-v1, and a small number of page refreshes display star ratings from reviews-v3. The result proves that the Istio traffic weight distribution rule takes effect normally.

```
PS C:\WINDOWS\system32\istio-1.23.1> kubectl apply -f samples/bookinfo/networking/virtual-service-reviews-80-20.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32\istio-1.23.1> _
```

his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

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Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

Reviewer1
Reviews served by: reviews-v1-dbf45bbb-ngbp8

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

Reviewer2
Reviews served by: reviews-v1-dbf45bbb-ngbp8

3.2 User-Based Routing Policy (Jason User Specific Routing)

Experiment Purpose: Configure exclusive routing rules to forward all access requests from the user Jason to the reviews-v3 service with star ratings, while traffic from other common users is randomly distributed to different service versions.

Operation Steps: Apply the user routing configuration file to bind exclusive access rules for the Jason user.

Experiment Result: After logging in to the page with the Jason account, the book review area consistently displays star ratings from the reviews-v3 service without random version switching. It verifies that Istio can accurately identify user identities and execute customized routing policies.

```
PS C:\WINDOWS\system32\istio-1.23.1> kubectl apply -f samples/bookinfo/networking/virtual-service-reviews-jason-v2-v3.yaml
virtualservice.networking.istio.io/reviews configured
PS C:\WINDOWS\system32\istio-1.23.1>
```

his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

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Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

Reviewer1
Reviews served by: reviews-v3-76bd4c4c9d-fnmqx

★★★★★ ☆

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

Reviewer2
Reviews served by: reviews-v3-76bd4c4c9d-fnmqx

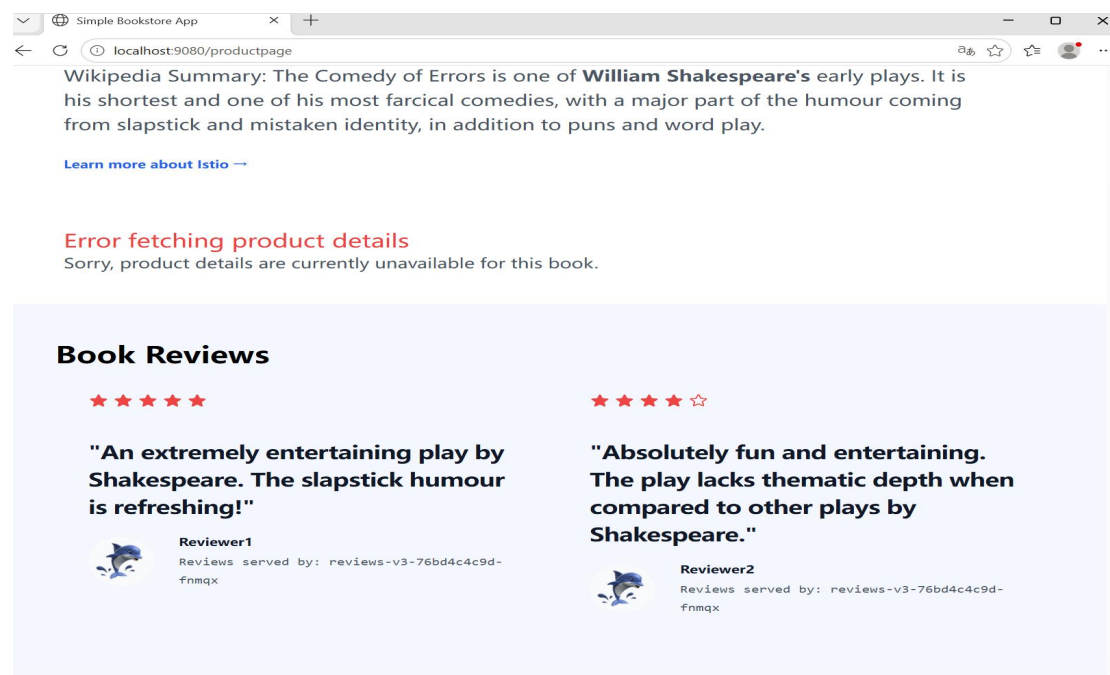
3.3 Service Fault Injection (Details Service)

Experiment Purpose: Inject artificial faults into the book details service to simulate abnormal service errors, and test the fault tolerance and service isolation capability of the microservice system.

Operation Steps: Apply the fault injection configuration file to set abnormal fault rules for the details service.

Experiment Result: The book details area on the page pops up an error message indicating that product details cannot be acquired and the details service is temporarily unavailable. Meanwhile, the book review area works normally without any abnormality. The result verifies that the fault is completely isolated to the details service and does not affect other associated services.

```
PS C:\WINDOWS\system32\istio-1.23.1> kubectl apply -f samples/bookinfo/networking/fault-injection-details-v1.yaml
Warning: virtualService rule #1 not used (only the last rule can have no matches)
virtualservice.networking.istio.io/details created
destinationrule.networking.istio.io/details created
PS C:\WINDOWS\system32\istio-1.23.1> _
```



4. Experiment Conclusion

This experiment successfully verifies three core traffic management capabilities of Istio in microservice architecture:

1. **Traffic Weight Distribution:** Accurately distribute request traffic among different versions of microservices to support smooth gray-scale release and traffic testing.
2. **User Customized Routing:** Implement differentiated traffic forwarding based on user identities to achieve precise and flexible traffic management.
3. **Artificial Fault Injection:** Actively simulate service failure scenarios to

effectively test the fault tolerance and stability of microservice systems.

4. **Service Fault Isolation:** Ensure that faults of a single service will not spread to other services, significantly improving the overall stability and anti-risk capability of the system.

The experimental results fully prove that Istio provides powerful and flexible traffic management solutions for microservice architecture. It can realize fine-grained traffic control, customized user access experience and system fault simulation, which effectively improves the operation and maintenance efficiency and operational reliability of microservice applications.